



Course Code: CSE475
Course Title: Machine Learning

Section: 06

[Semester: Summer 2026]

Project Task: 01
Project Title: Explainable ML model for Skin-Cancer Detection

Submitted by (Group-2):

Student Name	Student ID
Md. Rahat Khan	2021-1-60-090
Fahim Hossain	2022-2-60-142
Md Ahsiul Karim	2022-3-60-074
Jannatul Alam Shifa	2022-2-60-147

Submitted To:
Dr. Raihan Ul Islam
Associate Professor, Department of Computer Science and Engineering,
East West University

Related Work Table:

Title	Authors	Year	Dataset	Application area	Method/Model	Metrics	Strengths	Limitations	Research gap	Relation to your work
DSCC_Net: MultiClassification Deep Learning Models for Diagnosing Skin Cancer	Tahir et al	2023	ISIC 2020, HAM10000 and DermIS; four classes	Four-class skin cancer classification	Custom CNN, SMOTE-Tomek and Grad-CAM	Accuracy 94.17%; F1 93.93%; AUC 99.43%	High performance; class balancing and XAI	Mixed datasets; possible leakage; limited skin-tone diversity	External validation and patient-level splitting	Supports class balancing, CNN comparison and XAI
Deep LearningBased Skin Diseases Classification Using Smartphones	Oztel et al.	2023	PAD-UFE S-20 and Monkeypox Skin Lesion Dataset	Mobile skin disease screening	ResNet18, EfficientNet-B3 and TensorFlow Lite	ResNet18 accuracy 74.27%; EfficientNetB3 75.88%	Real Android deployment and speed testing	Moderate accuracy; no clinical validation	Improve mobile accuracy using hospital images	Useful for mobile deployment of the proposed model
Skin Cancer Classification Using Vision Transformers and XAI	Dagnaw et al.	2024	ISIC subset; 3,297 images	Benign versus malignant classification	ViT, Swin, CNN and Grad-CAM variants	ResNet50 accuracy 88.8%; F1 87.8%	Compares CNNs and transformers with XAI	Small dataset; qualitative XAI evaluation	Quantitative XAI and external testing	Provides CNN, ViT and Swin baselines
Enhancing Dermatological Diagnostics with EfficientNet	Manole et al.	2024	ISIC 2019 and private clinical images	Four-class and sixclass lesion classification	Customized EfficientNetB3	Four-class accuracy about 96%; six-class about 89%	Detailed class-wise evaluation	Private data; unclear split; weak SCC recall	More real minority-class images and external validation	Strong EfficientNet reference for the study
EfficientSkinDis	Rafay and Hussain	2023	Atlas Dermatology and ISIC; 31 classes	Multi-disease skin classification	EfficientNet-B2 with Streamlit deployment	Accuracy 87.15%; F1 0.87; AUC 0.91	Large class coverage and cross-validation	Source bias; no patient-level split	Clinically verified multi-source evaluation	Shows scalability and web deployment
Hybrid ViT-GNN with RegionAdaptive Attention	Dogga et al.	2026	ISIC 2020, HAM10000 and PH2	Skin cancer classification and XAI	ViT, GNN, RAA, Grad-CAM and SHAP	ISIC 94.3%; HAM10000 88.7%; PH2 89.5%	Combines global, spatial and attention features	Inconsistent results and limited reproducibility	Reproducible validation and real ablation studies	Relevant to hybrid transformer and graph-based work
Leveraging Spatial and Semantic Feature Extraction with Capsule Networks and GNNs	Santoso et al.	Not stated	HAM10000; 10,015 images and seven classes	Seven-class lesion classification	PViG-CapsNet-Tiny	Accuracy 95.52%; melanoma recall 0.692	Compact model and strong overall accuracy	Unclear split; no external validation; moderate melanoma recall	Patient-level testing and better minority-class evaluation	Most directly related HAM10000 graphbased benchmark